

Quantum Change Intervals: Exact Asymptotic Localization with Collective Measurements

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We consider exact minimum-error localization of a transient change in a sequence of independent pure-state outputs. The source emits $|0\rangle$, switches abruptly to a calibrated state $|\psi\rangle$ on one nonempty contiguous interval, and then returns perfectly to $|0\rangle$. The outputs may be stored coherently and measured collectively. For a known interval length i , the candidate-state Gram matrix is $G_{ab} = r^{\min(|a-b|, i)}$, where $c = |\langle 0|\psi\rangle|$ and $r = c^2$. For each fixed i , its Toeplitz symbol gives an exact integral for the asymptotic Bayes success probability, and the square-root measurement (SRM) attains the same limit. When both the known length and the number of admissible translations diverge, with no restriction on their ratio, the optimal success probability converges to the one-dimensional Toeplitz (single-change) functional evaluated at the effective compound overlap c^2 , namely $p_1(c^2)$. For an unknown length, the physical Gram kernel is not globally two-dimensional Toeplitz because separated intervals carry a gap-dependent correction. Under the prior uniform over all nonempty intervals, we remove a vanishing-prior short sector, use a triangular Følner reduction, and transfer the result back to both the Bayes optimum and the SRM of the full physical family. An exceptional-sector Gram transfer theorem isolates the reusable mechanism behind this step. The resulting, prior-specific limit is $p_1(c)^2$. For the stated conditional uniform anomaly prior, adding a no-change hypothesis with fixed prior changes the optimal joint Bayes limit to $\pi_0 + (1 - \pi_0)L$, where L is the conditional localization limit; no augmented weighted-SRM claim is made. Finite-size semidefinite programs and full dense physical-Gram SRM computations illustrate the asymptotic results.

Keywords: quantum change point; quantum state discrimination; change interval; square-root measurement; Toeplitz operator; Følner sequence; Bayesian detection

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1 Introduction

Quantum change-point inference asks where an ordered source changes from one state-generating process to another. Unlike a classical observer, the receiver may retain the systems and measure the entire sequence collectively. Early Bayesian work considered unknown pure states and two possible change locations [1]. For a single permanent change, collective minimum-error discrimination has a nonzero asymptotic success probability and can outperform online local strategies [2]; exact and online unambiguous-identification protocols were subsequently developed in Refs. [3, 4]. An adaptive Bayesian online protocol was also demonstrated experimentally with photonic state sequences [5]. Separate formulations optimize different operational risks. Quickest and sequential quantum change detection trade false-alarm control against detection delay [6, 7]. Hamiltonian change points are finite-candidate quantum process-discrimination problems under a minimum-error criterion [8], whereas the 2026 channel result studies unambiguous process discrimination and hence permits an inconclusive outcome [9]. The literature also includes general Gram-matrix semidefinite programs, multiple changes, isolated anomalies, and boundaries between quantum domains [10–13].

Here the change is transient: the source returns to its reference state after one nonempty contiguous anomalous block. We refer to this interval-constrained family as a *quantum change interval*. For a sequence of length n and an interval $I = [a, b]$, the corresponding hypothesis is

$$|\Phi_{a,b}\rangle = |0\rangle^{\otimes(a-1)}|\psi\rangle^{\otimes(b-a+1)}|0\rangle^{\otimes(n-b)}.$$

When the length $i = b - a + 1$ is known, only the translation a is inferred; when the length is unknown, both endpoints are inferred. Figure 1 summarizes this returning geometry and the collective exact-localization task.

The model is related to established quantum tasks through exact restrictions. The boundary slice $b = n$ is the permanent quantum change-point family [2–4]. In the two-change notation of Mohan–Sikora–Upadhyay [10], the present candidates are obtained by setting $N = n$, $c_1 = a - 1$, $c_2 = b$, and choos-

ing the initial, middle, and final states as $|0\rangle, |\psi\rangle, |0\rangle$, respectively. After the phase convention used below, their three pairwise overlaps specialize to $c, c, 1$. Thus their piecewise two-change Gram kernel contains the returning family, while the nonempty-interval restriction imposes $c_1 < c_2$. For fixed length i , the interval candidates are the consecutive $k = i$ subensemble of multi-anomaly detection [12]; the containment is strict for $1 < i < n$, whereas the hypothesis sets coincide at $i = 1$ and $i = n$. The unknown-length family is the union of the consecutive subensembles. Quantum edge detection locates one boundary between unknown pure-state domains and proves an SRM-based protocol asymptotically optimal [13]. It is a single-boundary geometric analogue: the present model instead ties two endpoints to one interval, uses calibrated local states, and fixes both exterior domains to $|0\rangle$.

The unknown-length problem has a specific obstruction that is hidden by the two-endpoint notation. For intersecting intervals, the overlap kernel depends only on the endpoint displacement and agrees with a separable two-dimensional Toeplitz kernel. For separated intervals, however, the physical overlap contains an additional correction determined by the gap between the two intervals. Thus the exact physical Gram matrix is neither a global two-dimensional Toeplitz compression nor an element of a transitive association scheme. A direct multilevel Szegő theorem therefore does not solve the problem. The proof deletes a short-interval sector whose prior mass vanishes, compares the retained long-interval Gram matrix in operator norm with a separable convolution kernel on a triangular Følner sequence, and obtains matching SRM and optimum limits through local-diagonal functional calculus and a Gram upper bound. A prior-mass sandwich transfers the optimum to all intervals, while principal-submatrix interlacing, including the essential retained-family factor, transfers the square-root trace and hence the SRM of the full physical family.

Throughout this paper, *exact localization* refers to the zero-one reward: an outcome is successful only when the reported interval $\hat{I}$ equals the true interval I . The minimum-error POVM reports an interval on every run and may err. In unambiguous exact identification, every conclusive report must be correct and an inconclusive outcome is allowed [3, 12]. Certified-answer discrimination instead bounds the distance between a reported and true ordered label, interpolating between unambiguous and minimum-error criteria [14]. We study the minimum-error zero-one reward, including its joint extension with a no-anomaly label.

One operational scenario considers a device emitting a batch of quantum probes or carriers during a transient coherent disturbance. The model assumes independent single-system outputs, a calibrated pure anomalous state $|\psi\rangle$, an abrupt switch from $|0\rangle$, a state that remains constant throughout the anomalous

block, and a perfect return to $|0\rangle$. A receiver may retain the complete batch coherently and apply an unrestricted global POVM to estimate the onset and duration. We treat this as an ultimate collective-measurement benchmark. Correlated or mixed outputs, gradual switching, imperfect recovery, unknown fault states, and restricted storage or readout architectures require different models.

The same returning geometry appears in classical statistics. Epidemic alternatives and changed-segment models place a temporary distributional change on an unknown contiguous interval [15, 16], while scan methods cover spatial or graph-structured changed regions [17, 18]. Those works optimize classical test statistics and type-I-error or power criteria. Here the observations are noncommuting quantum hypotheses, arbitrary collective POVMs are allowed, and the target is the exact Bayes localization probability. Table 1 summarizes the resulting distinctions.

We use *quantum change interval* as a descriptive name for this returning contiguous ensemble, a structured instance of multiple quantum changes and anomalous-site identification. Toeplitz and circulant asymptotics, including multilevel spectral-distribution theorems, provide the matrix-analytic background [19–21]. The contribution is the combination of exact asymptotic Bayes laws, a Toeplitz/Følner reduction for the non-Toeplitz physical kernel, and an exceptional-sector Gram theorem that transfers both the optimum and SRM limit to the full physical family. This transfer mechanism applies to other structured Gram ensembles satisfying its bulk-comparison hypotheses. More precisely:

1. For fixed known length i , the optimal and SRM success probabilities converge to a one-dimensional Toeplitz integral, with an $O(N^{-1/2})$ bound on their difference.
2. If $i = i_n \rightarrow \infty$ and the number $N_n = n - i_n + 1$ of admissible translations also diverges, both probabilities converge to $p_1(c^2)$, with no assumption on i_n/N_n .
3. Under the prior uniform on all nonempty intervals, both probabilities converge to the prior-specific law $p_1(c)^2$. The change from $p_1(c^2)$ to $p_1(c)^2$ reflects the replacement of one translation coordinate by two endpoint coordinates together with the elementary overlap change from c^2 to c .
4. If the no-anomaly prior $\pi_0 \in (0, 1)$ is fixed and the conditional localization limit is L , the optimal joint detection-and-localization success probability converges to $\pi_0 + (1 - \pi_0)L$. This item makes no augmented weighted-SRM claim.

All results assume independent outputs, abrupt switching, a calibrated pure anomalous state, and a perfect return to $|0\rangle$. The overlap c is fixed as $n \rightarrow \infty$, and π_0 is fixed whenever H_0 is present. The stated uniform priors and arbitrary collective POVMs define

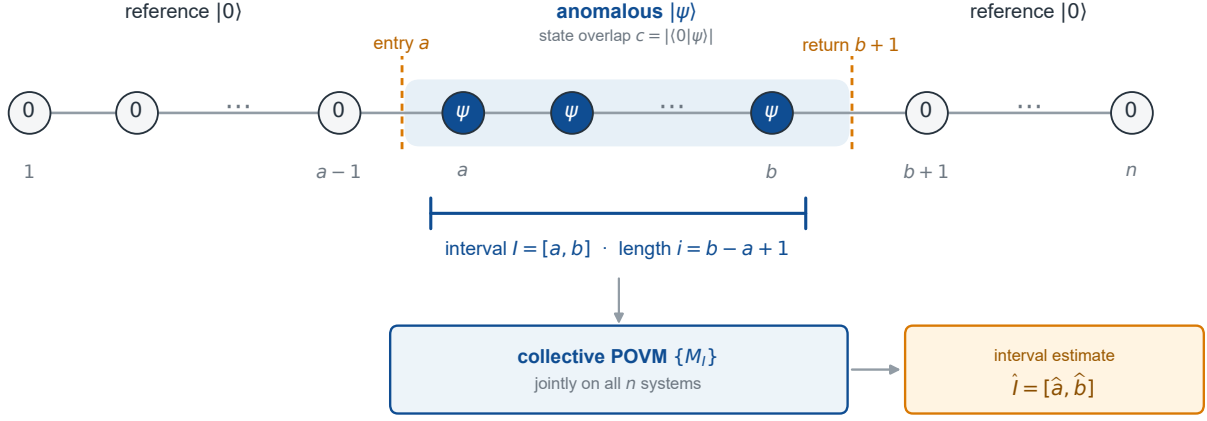


Figure 1: **Returning quantum change interval and collective localization.** A source emits the reference state $|0\rangle$ except on a nonempty contiguous interval $I = [a, b]$, where it emits the known anomalous state $|\psi\rangle$. The entry and return boundaries are a and $b + 1$, respectively. For known length, $i = b - a + 1$ is fixed and only a is inferred; for unknown length, both endpoints are inferred. An arbitrary collective POVM may act on all n systems and returns an exact interval estimate $\hat{I}$. The state overlap is $c = |\langle 0|\psi\rangle|$; for $0 < c < 1$, the local states are nonorthogonal.

Table 1: Theorem-level comparison and structural relations between quantum change intervals and neighboring quantum and classical models.

Work	Exact relation	Established result and present delta
Sentís et al. [2–4]	Exactly the boundary slice $b = n$	Permanent-change minimum-error limit, with separate unambiguous and online protocols; a returning interval couples entry and return boundaries.
Mohan–Sikora–Upadhyay [10]	Two-change specialization $c_1 = a - 1, c_2 = b$, with states $ 0\rangle, \psi\rangle, 0\rangle$	General Gram-only Bayes SDP and piecewise two-change kernel; no closed returning-interval asymptotics or full-family SRM transfer.
Quantum multi-anomaly detection [12]	Consecutive $k = i$ subensemble; strict for $1 < i < n$	Johnson symmetry gives exact finite- n SRM optimality. The exact zero-error law $(1 - c^2)^k$ has a local implementation and is also the fixed- k minimum-error limit; contiguity destroys Johnson transitivity.
Quantum edge detection [13]	Single-boundary geometric analogue with unknown domains	Its SRM protocol is asymptotically optimal; here two tied endpoints, known states, and equal exterior domains require a non-Toeplitz-to-Følner transfer.
Epidemic/changed-interval statistics [15–18]	Geometric analogue of a returning interval	Classical scan/CUSUM detection uses significance and power criteria; the present task optimizes exact Bayes localization over collective POVMs.

the decision problems. Section 8 discusses extensions beyond one interval, fixed states and overlaps, and unrestricted coherent readout.

2 Model and discrimination preliminaries

2.1 Phase convention and Gram matrices

Write

$$\langle 0|\psi\rangle = ce^{i\varphi}, \quad 0 \leq c \leq 1, \quad r = c^2.$$

Replacing $|\psi\rangle$ by $e^{-i\varphi}|\psi\rangle$ changes each candidate vector only by a hypothesis-dependent global phase. Equivalently, the original and phase-fixed Gram matrices are related by a diagonal unitary conjugation. Minimum-error and SRM success probabilities are unchanged. We therefore use

$$\langle 0|\psi\rangle = c \geq 0$$

without loss of generality.

Unless a theorem explicitly introduces a sequence $i = i_n$, every asymptotic statement is taken as $n \rightarrow \infty$ with c (and hence $r = c^2$) fixed. In Section 6, the no-anomaly prior $\pi_0 \in (0, 1)$ is also fixed rather than scaled with n .

For an ensemble of M normalized pure states $\{|\phi_j\rangle\}_{j=1}^M$ with equal priors, let

$$G_{jk} = \langle \phi_j | \phi_k \rangle.$$

We write $P_{\text{opt}}(G)$ for the minimum-error optimum over all POVMs [22–25] and

$$P_{\text{SRM}}(G) = \frac{1}{M} \sum_{j=1}^M (\sqrt{G})_{jj}^2$$

for the square-root-measurement success probability [26–28]. If G is singular, $G^{-1/2}$ is understood on the support of G , and the POVM is completed arbitrarily on its orthogonal complement; this does not change any ensemble success probability. The elementary trace bound

$$\left(\frac{\text{tr} \sqrt{G}}{M}\right)^2 \leq P_{\text{SRM}}(G) \leq P_{\text{opt}}(G) \quad (2.1)$$

will be used repeatedly.

2.2 Stability under Gram perturbations

Lemma 2.1 (Operator-norm Gram stability). *Let $G, H \in \mathbb{C}^{M \times M}$ be positive semidefinite with unit diagonal. Then*

$$|P_{\text{opt}}(G) - P_{\text{opt}}(H)| \leq \|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq \|G - H\|_{\text{op}}^{1/2} \quad (2.2)$$

and

$$\begin{aligned} |P_{\text{SRM}}(G) - P_{\text{SRM}}(H)| \\ \leq 2\|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq 2\|G - H\|_{\text{op}}^{1/2}. \end{aligned} \quad (2.3)$$

If $G, H \geq \mu I$ for some $\mu > 0$, then

$$\|\sqrt{G} - \sqrt{H}\|_{\text{op}} \leq \frac{\|G - H\|_{\text{op}}}{2\sqrt{\mu}}. \quad (2.4)$$

Proof. Use the canonical Gram realizations $|x_j\rangle = \sqrt{G}e_j$ and $|y_j\rangle = \sqrt{H}e_j$. Their norms are one because the Gram matrices have unit diagonal, and the trace distance between the corresponding pure states is at most $\|x_j - y_j\|$. Apply an optimal POVM for either ensemble to the other and average over j to obtain the first inequality in (2.2). The second is the operator Hölder inequality for positive square roots. Since $0 \leq (\sqrt{G})_{jj}, (\sqrt{H})_{jj} \leq 1$, the scalar bound $|a^2 - b^2| \leq 2|a - b|$ gives (2.3). For the final claim, $X = \sqrt{G} - \sqrt{H}$ solves

$$\sqrt{G}X + X\sqrt{H} = G - H.$$

The inverse Sylvester operator has norm at most $(2\sqrt{\mu})^{-1}$, which proves (2.4). $\square$

3 A known interval length

Let the anomalous interval have known length $i \geq 1$. There are

$$N = n - i + 1$$

admissible starting points, and the equal-prior candidates are

$$\begin{aligned} |\Phi_a^{(i)}\rangle &= |0\rangle^{\otimes(a-1)} |\psi\rangle^{\otimes i} |0\rangle^{\otimes(n-a-i+1)}, \\ a &= 1, \dots, N. \end{aligned}$$

Their exact Gram matrix is

$$(G_{N,i})_{ab} = r^{\min(|a-b|, i)}. \quad (3.1)$$

Define

$$f_{i,r}(\theta) = 1 - r^i + 2 \sum_{d=1}^{i-1} (r^d - r^i) \cos(d\theta). \quad (3.2)$$

3.1 Fixed-length limit

Theorem 3.1 (Fixed-length localization). *For every fixed integer $i \geq 1$ and $0 \leq r \leq 1$,*

$$\lim_{N \rightarrow \infty} P_{\text{opt}}(G_{N,i}) = \lim_{N \rightarrow \infty} P_{\text{SRM}}(G_{N,i}) = P_{\infty}(i, r), \quad (3.3)$$

where

$$P_{\infty}(i, r) = \left[\frac{1}{2\pi} \int_0^{2\pi} \sqrt{f_{i,r}(\theta)} d\theta \right]^2. \quad (3.4)$$

For fixed i and $0 < r < 1$,

$$0 \leq P_{\text{opt}}(G_{N,i}) - P_{\text{SRM}}(G_{N,i}) = O_{i,r}(N^{-1/2}). \quad (3.5)$$

Proof. Put $q = r^i$ and decompose

$$G_{N,i} = qJ_N + B_{N,i},$$

where $B_{N,i} = T_N(f_{i,r})$ has entries

$$(B_{N,i})_{ab} = \begin{cases} r^{|a-b|} - r^i, & |a-b| < i, \\ 0, & |a-b| \geq i. \end{cases}$$

Choose a local basis with $|\psi\rangle = \sqrt{r}|0\rangle + \sqrt{1-r}|1\rangle$. After removing the common all-zero component, each candidate has a basis component, with squared amplitude $(1-r)^i$, that occurs in no other candidate. Hence

$$B_{N,i} \geq (1-r)^i I_N. \quad (3.6)$$

The corresponding infinite residual convolution operator has symbol $f_{i,r}$, so (3.6), applied to arbitrarily long finitely supported vectors, implies

$$f_{i,r}(\theta) \geq (1-r)^i. \quad (3.7)$$

For $N \geq 2i$, let $\tilde{B}_{N,i}$ be the circulant matrix with the same local coefficients and set

$$C_{N,i} = r^i J_N + \tilde{B}_{N,i}.$$

Its eigenvalues away from the uniform Fourier mode are the sampled values $f_{i,r}(2\pi k/N)$; the uniform

mode receives the additional positive rank-one contribution. Thus $C_{N,i}$ is a positive definite, unit-diagonal circulant Gram matrix. The matrices $B_{N,i}$ and $\tilde{B}_{N,i}$ differ only in the upper-right and lower-left wrap-around corner blocks.

To make the boundary estimate explicit, set

$$b_d = r^d - r^i, \\ E_{N,d} = \sum_{a=1}^d e_a e_{N-d+a}^*, \quad 1 \leq d < i.$$

Here $E_{N,d}$ is a rank- d partial permutation from the last d coordinates to the first d coordinates, and

$$\tilde{B}_{N,i} - B_{N,i} = \sum_{d=1}^{i-1} b_d (E_{N,d} + E_{N,d}^*). \quad (3.8)$$

For $N \geq 2i$, the initial and terminal coordinate subspaces in (3.8) are disjoint. Consequently $\|E_{N,d} + E_{N,d}^*\|_1 = 2d$. Moreover, the whole difference maps between the span of the first $i-1$ coordinates and the span of the last $i-1$ coordinates. The triangle inequality and this two-corner support therefore give

$$\|G_{N,i} - C_{N,i}\|_1 \leq K_{i,r} := 2 \sum_{d=1}^{i-1} d(r^d - r^i), \quad (3.9)$$

$$\text{rank}(G_{N,i} - C_{N,i}) \leq 2(i-1). \quad (3.10)$$

The Sylvester integral gives

$$\|\sqrt{G_{N,i}} - \sqrt{C_{N,i}}\|_1 \leq \frac{K_{i,r}}{2(1-r)^{i/2}} = O_{i,r}(1). \quad (3.11)$$

Because $C_{N,i}$ and $\sqrt{C_{N,i}}$ are circulant, the diagonal of $\sqrt{C_{N,i}}$ is constant. If

$$\beta_N = \frac{1}{N} \text{tr} \sqrt{C_{N,i}},$$

then the SRM value is β_N^2 . The Yuen–Kennedy–Lax dual operator

$$Y = \frac{\beta_N}{N} \sqrt{C_{N,i}}$$

is feasible because, with $S = \sqrt{C_{N,i}}$,

$$\beta_N S - S e_j e_j^* S \geq 0.$$

Indeed, since $S \succ 0$, congruence by $S^{-1/2}$ reduces this to

$$\beta_N I - S^{1/2} e_j e_j^* S^{1/2} \geq 0.$$

The rank-one term has unique nonzero eigenvalue $e_j^* S e_j = \beta_N$, because the diagonal of S is constant. Since $\text{tr} Y = \beta_N^2$, the SRM is exactly optimal for the circulant model.

Let

$$\tau_N = \|\sqrt{G_{N,i}} - \sqrt{C_{N,i}}\|_1.$$

Put $\Delta_N = \sqrt{G_{N,i}} - \sqrt{C_{N,i}}$. The canonical Gram realizations, followed by Cauchy–Schwarz and the monotonicity of Schatten norms, give the explicit chain

$$\begin{aligned} |P_{\text{opt}}(G_{N,i}) - P_{\text{opt}}(C_{N,i})| &\leq \frac{1}{N} \sum_{j=1}^N \|\Delta_N e_j\|_2 \\ &\leq \frac{1}{\sqrt{N}} \left(\sum_{j=1}^N \|\Delta_N e_j\|_2^2 \right)^{1/2} \\ &= \frac{\|\Delta_N\|_{\text{HS}}}{\sqrt{N}} \\ &\leq \frac{\|\Delta_N\|_1}{\sqrt{N}} = \frac{\tau_N}{\sqrt{N}}. \end{aligned} \quad (3.12)$$

while the SRM diagonal formula gives

$$|P_{\text{SRM}}(G_{N,i}) - P_{\text{SRM}}(C_{N,i})| \leq \frac{2\tau_N}{N}.$$

Together with (3.11), this proves (3.5). Finally, the sampled eigenvalues form a Riemann sum. The rank-one uniform-mode correction changes $N^{-1} \text{tr} \sqrt{C}$ by at most $O(N^{-1/2})$, and therefore

$$\frac{1}{N} \text{tr} \sqrt{C_{N,i}} \longrightarrow \frac{1}{2\pi} \int_0^{2\pi} \sqrt{f_{i,r}(\theta)} d\theta.$$

This proves (3.3)–(3.4) for $0 < r < 1$. The cases $r = 0, 1$ are handled below. $\square$

Corollary 3.2 (Residual Toeplitz family). *Fix $i \geq 1$ and $0 \leq r < 1$, put $q = r^i$, and let*

$$R_{N,i} := \frac{G_{N,i} - qJ_N}{1-q} = T_N \left(\frac{f_{i,r}}{1-r^i} \right).$$

Then $R_{N,i}$ is the Gram matrix of normalized residual states and

$$\lim_{N \rightarrow \infty} P_{\text{opt}}(R_{N,i}) = \lim_{N \rightarrow \infty} P_{\text{SRM}}(R_{N,i}) = \frac{P_{\infty}(i,r)}{1-r^i}. \quad (3.13)$$

Equivalently, $(1-r^i)P_{\text{opt}}(R_{N,i}) \rightarrow P_{\infty}(i,r)$, and the same statement holds with P_{SRM} .

Proof. The diagonal of $G_{N,i} - qJ_N$ is $1-q$, and (3.6) shows that it is positive definite. Thus $R_{N,i}$ is a unit-diagonal Gram matrix. Its Toeplitz symbol is $h_{i,r} = f_{i,r}/(1-r^i)$, which is bounded above and below away from zero. The one-dimensional instance of Lemma 5.1, followed by Lemma 5.2 and (2.1), gives the common limit

$$\left[\frac{1}{2\pi} \int_0^{2\pi} \sqrt{h_{i,r}(\theta)} d\theta \right]^2 = \frac{P_{\infty}(i,r)}{1-r^i}.$$

$\square$

3.2 Exact and special-function cases

Corollary 3.3 (One-site interval, finite size). *For $i = 1$,*

$$G_{N,1} = (1-r)I_N + rJ_N$$

is permutation invariant. Its positive square root has constant diagonal, so the preceding dual certificate proves finite-size SRM optimality. Since its eigenvalues are $1 + (N-1)r$ once and $1-r$ with multiplicity $N-1$,

$$\begin{aligned} P_{\text{opt}}(G_{N,1}) &= P_{\text{SRM}}(G_{N,1}) \\ &= \frac{[\sqrt{1+(N-1)r} + (N-1)\sqrt{1-r}]^2}{N^2}. \end{aligned} \quad (3.14)$$

In particular, $P_{\infty}(1, r) = 1 - r$.

This is the $k = 1$ specialization of the known single-anomaly result in Refs. [11, 12], after identifying their anomaly overlap parameter with $r = c^2$. Its inclusion records how the interval family recovers the established finite-size formula at $i = 1$.

Corollary 3.4 (Two-site interval). *With*

$$E(m) = \int_0^{\pi/2} \sqrt{1 - m \sin^2 u} \, du,$$

one has

$$P_{\infty}(2, r) = \frac{4}{\pi^2} (1-r)(1+3r) E^2\left(\frac{4r}{1+3r}\right). \quad (3.15)$$

Derivation. Equation (3.2) becomes

$$\begin{aligned} f_{2,r}(\theta) &= (1-r)(1+r+2r \cos \theta) \\ &= (1-r)(1+3r-4r \sin^2(\theta/2)). \end{aligned}$$

With $u = \theta/2$ and symmetry about $\pi/2$,

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} \sqrt{f_{2,r}(\theta)} \, d\theta \\ = \frac{2}{\pi} \sqrt{(1-r)(1+3r)} E\left(\frac{4r}{1+3r}\right). \end{aligned}$$

Squaring proves (3.15). $\square$

3.3 Endpoint parameters

If $r = 0$, then $G_{N,i} = I_N$ and both success probabilities equal one. If $r = 1$, then $G_{N,i} = J_N$, all hypotheses represent the same ray, and both probabilities equal $1/N$. These values agree with the continuous limiting formulas. All spectral-gap arguments above are used only for $0 < r < 1$.

4 A growing known length

Define the Poisson kernel

$$g_r(\theta) = \sum_{d \in \mathbb{Z}} r^{|d|} e^{id\theta} = \frac{1-r^2}{1-2r \cos \theta + r^2}$$

and

$$p_1(r) = \left[\frac{1}{2\pi} \int_0^{2\pi} \sqrt{g_r(\theta)} \, d\theta \right]^2 = \frac{4(1-r^2)}{\pi^2} K^2(r^2), \quad (4.1)$$

where K is the complete elliptic integral of the first kind in the parameter convention and $p_1(1) = 0$ by continuity. The last equality follows from the standard Landen transformation; the integral representation is the definition used in all proofs.

Proposition 4.1 (One-change Toeplitz benchmark). *For*

$$Q_N = (r^{|a-b|})_{a,b=1}^N, \quad 0 \leq r < 1,$$

one has

$$\begin{aligned} P_{\text{opt}}(Q_N), P_{\text{SRM}}(Q_N) &\longrightarrow p_1(r), \\ \frac{1}{N} \text{tr} \sqrt{Q_N} &\longrightarrow \sqrt{p_1(r)}. \end{aligned} \quad (4.2)$$

Proof. The matrix Q_N is the compression of the one-dimensional convolution operator with kernel $r^{|d|}$, whose symbol is g_r and is bounded above and below away from zero. Lemma 5.1, with $d = 1$ and $\phi(t) = \sqrt{t}$, gives convergence in mean of the square-root diagonal to $(2\pi)^{-1} \int \sqrt{g_r} = \sqrt{p_1(r)}$. Lemma 5.2 supplies the matching minimum-error upper bound, while (2.1) supplies the lower bound and the SRM squeeze. This also recovers the benchmark of Ref. [2]. $\square$

Theorem 4.2 (Long known interval). *Let $i = i_n$ and $N_n = n - i_n + 1$. If*

$$i_n \rightarrow \infty, \quad N_n \rightarrow \infty, \quad (4.3)$$

then, for every fixed $0 \leq r \leq 1$,

$$P_{\text{opt}}(G_{N_n, i_n}), P_{\text{SRM}}(G_{N_n, i_n}) \longrightarrow p_1(r) = p_1(r^2). \quad (4.4)$$

No condition on i_n/N_n is required. The convergence is pointwise in fixed r . No finite-size rate or uniformity as $r \uparrow 1$ is claimed.

Proof. The endpoint values $r = 0, 1$ follow from the matrices described in Section 3.3. We therefore assume $0 < r < 1$ and put $q_i = r^i$. Separating the common all-zero component gives

$$|\Phi_a^{(i)}\rangle = \sqrt{q_i} |\Omega\rangle + \sqrt{1-q_i} |\xi_a^{(i)}\rangle, \quad \langle \Omega | \xi_a^{(i)} \rangle = 0.$$

The normalized residual Gram matrix is

$$(R_{N,i})_{ab} = \begin{cases} \frac{r^{|a-b|} - r^i}{1 - r^i}, & |a-b| < i, \\ 0, & |a-b| \geq i. \end{cases} \quad (4.5)$$

Its symbol

$$h_{i,r}(\theta) = 1 + 2 \sum_{d=1}^{i-1} \frac{r^d - r^i}{1 - r^i} \cos(d\theta)$$

satisfies

$$\|h_{i,r} - g_r\|_\infty \leq \frac{2(i-1)r^i}{1 - r^i} + \frac{2r^i}{1 - r} =: \varepsilon_i(r) \rightarrow 0. \quad (4.6)$$

Let $Q_N = (r^{|a-b|})_{a,b=1}^N$. Toeplitz compression gives

$$\|R_{N,i} - Q_N\|_{\text{op}} \leq \varepsilon_i(r). \quad (4.7)$$

Since $g_r \geq (1-r)/(1+r) > 0$, both matrices have a common positive spectral gap for large i . Lemma 2.1 therefore transfers the minimum-error optimum uniformly in N . The trace distance between each original state and its residual state is also $\sqrt{q_i} = r^{i/2}$, so

$$|P_{\text{opt}}(G_{N,i}) - P_{\text{opt}}(R_{N,i})| \leq r^{i/2}.$$

The single-change-point Toeplitz limit for Q_N therefore proves

$$P_{\text{opt}}(G_{N,i}) \rightarrow p_1(r).$$

For the SRM, the rank-one common component requires a separate trace estimate. Since

$$G_{N,i} = q_i J_N + (1 - q_i) R_{N,i},$$

operator monotonicity and the Rotfel'd trace inequality give the exact sandwich

$$\begin{aligned} \sqrt{1 - q_i} \operatorname{tr} \sqrt{R_{N,i}} &\leq \operatorname{tr} \sqrt{G_{N,i}} \\ &\leq \sqrt{1 - q_i} \operatorname{tr} \sqrt{R_{N,i}} + \sqrt{q_i N}. \end{aligned} \quad (4.8)$$

Equation (4.7), the common spectral gap, and the Toeplitz limit imply

$$\frac{1}{N} \operatorname{tr} \sqrt{R_{N,i}} \rightarrow \sqrt{p_1(r)}.$$

After dividing (4.8) by N , the remaining rank-one term is $\sqrt{q_i}/N \rightarrow 0$. Thus

$$\frac{1}{N} \operatorname{tr} \sqrt{G_{N,i}} \rightarrow \sqrt{p_1(r)}.$$

The sandwich (2.1) completes the SRM proof. $\square$

Remark 4.3 (An exact finite-size regime). If $i \geq N-1$, then $\min(|a-b|, i) = |a-b|$ for every matrix entry, and hence

$$G_{N,i} = Q_N$$

exactly. The approximation in Theorem 4.2 is needed only when $i < N-1$.

5 Unknown interval length

Let

$$\begin{aligned} \mathcal{I}_n &= \{[a, b] : 1 \leq a \leq b \leq n\}, \\ M_n &= |\mathcal{I}_n| = \frac{n(n+1)}{2}, \end{aligned}$$

and equip $\mathcal{I}_n$ with the uniform prior $1/M_n$. This is uniform over intervals, not over lengths. If $L = |I|$, the induced marginal is

$$\begin{aligned} \Pr\{L = \ell\} &= \frac{n - \ell + 1}{M_n} \\ &= \frac{2(n - \ell + 1)}{n(n+1)}, \quad 1 \leq \ell \leq n. \end{aligned} \quad (5.1)$$

Conditionally on $L = \ell$, the starting point is uniform over its $n - \ell + 1$ admissible values. Consequently $\mathbb{E}L = (n+2)/3$, and L/n converges to the triangular density $2(1-x)$ on $0 < x < 1$. Thus the prior favors shorter lengths, but its typical interval still has order- n length. The candidate indexed by $I = [a, b]$ has $|\psi\rangle$ on I and $|0\rangle$ elsewhere. Its exact Gram kernel is

$$G_{I,J}^{(n)} = c^{|I \triangle J|}. \quad (5.2)$$

Theorem 5.5 is therefore a Bayes law for the uniform-over-intervals prior in (5.1). Alternative priors may retain nonvanishing mass on bounded or logarithmically short intervals and lead to different limiting reductions.

For intersecting intervals,

$$|I \triangle J| = |a - a'| + |b - b'|. \quad (5.3)$$

For disjoint intervals, assume without loss of generality that $b < a'$ and define the separating gap by

$$g(I, J) := a' - b - 1 \geq 0.$$

Then the exact identity is

$$|a - a'| + |b - b'| = |I \triangle J| + 2g(I, J). \quad (5.4)$$

Thus (5.3) still holds for adjacent intervals, where $g(I, J) = 0$, but fails by twice the separating gap when $g(I, J) > 0$. The exact physical kernel is therefore not globally a two-dimensional Toeplitz kernel.

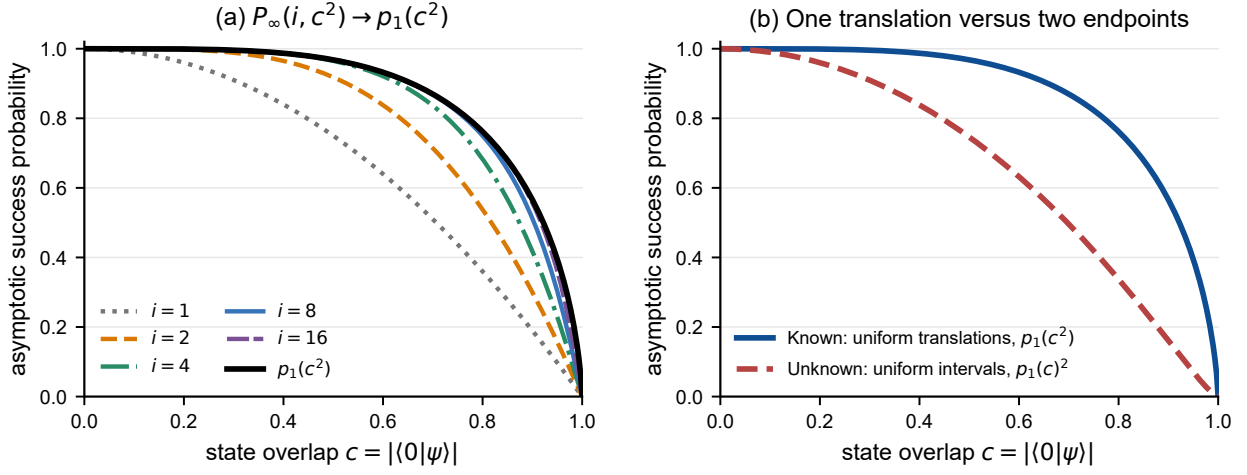


Figure 2: **Hierarchy of analytic localization limits.** (a) The fixed-length Toeplitz integral $P_\infty(i, c^2)$ approaches the long-known-interval law $p_1(c^2)$ as i grows. The $i = 2$ curve is equivalently given by the complete-elliptic-integral expression in Corollary 3.4. (b) The one-translation law $p_1(c^2)$ and two-endpoint law $p_1(c)^2$ are shown for their respective uniform-translation and uniform-interval priors. The latter follows from the two-dimensional Følner limit and separable comparison symbol.

Classical multilevel Szegő theorems describe spectral averages for genuine multilevel Toeplitz compressions [20, 21]. Here the triangular endpoint sets and, more importantly, the non-Toeplitz physical kernel require the following local-diagonal Følner statement together with a separate perturbative transfer back to $G^{(n)}$.

5.1 A Følner functional-calculus lemma

Lemma 5.1 (Local diagonal Følner limit). *Let $k \in \ell^1(\mathbb{Z}^d)$ satisfy $k(-u) = \overline{k(u)}$, let A be convolution by k on $\ell^2(\mathbb{Z}^d)$, and suppose its Fourier symbol satisfies*

$$0 < \mu \leq F(\theta) \leq L < \infty$$

almost everywhere. If $\Lambda_n \subset \mathbb{Z}^d$ is any finite Følner sequence and $A_n = P_n A P_n|_{\ell^2(\Lambda_n)}$, then for every continuous $\phi : [\mu, L] \rightarrow \mathbb{C}$,

$$\frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} |\langle \delta_x, \phi(A_n) \delta_x \rangle - a_\phi| \rightarrow 0, \quad (5.5)$$

where

$$a_\phi = \frac{1}{(2\pi)^d} \int_{[0, 2\pi]^d} \phi(F(\theta)) d\theta.$$

Proof. Since A has spectrum in $[\mu, L]$, the same is true of every compression A_n . Fix $\varepsilon > 0$ and put $\eta = \varepsilon/8$. By the Weierstrass theorem, applied to the real and imaginary parts if necessary, choose a polynomial

$$p(t) = \sum_{m=0}^s a_m t^m \quad \text{such that} \quad \sup_{t \in [\mu, L]} |\phi(t) - p(t)| < \eta.$$

Let $k_R = k \mathbf{1}_{\{\|u\|_\infty \leq R\}}$, let A_R be convolution by k_R , and set $A_{R,n} = P_n A_R P_n|_{\ell^2(\Lambda_n)}$. Young's inequality gives

$$\|A - A_R\|_{\text{op}}, \|A_n - A_{R,n}\|_{\text{op}} \leq \|k - k_R\|_{\ell^1}. \quad (5.6)$$

Choose R so large that the right-hand side of (5.6) is at most 1 and

$$C_p \|k - k_R\|_{\ell^1} < \eta, \quad C_p := \sum_{m=1}^s m |a_m| (L+1)^{m-1},$$

where the second condition is void if $C_p = 0$. The telescoping identity for $X^m - Y^m$, together with $\|A\|, \|A_n\| \leq L$ and $\|A_R\|, \|A_{R,n}\| \leq L+1$, then yields

$$\|p(A) - p(A_R)\|_{\text{op}}, \|p(A_n) - p(A_{R,n})\|_{\text{op}} < \eta. \quad (5.7)$$

Define

$$Q_t = \{h \in \mathbb{Z}^d : \|h\|_\infty \leq t\}$$

and the sR -interior and boundary by

$$\Lambda_n^{\circ, sR} = \{x \in \Lambda_n : x + Q_{sR} \subseteq \Lambda_n\}, \\ \partial_{sR} \Lambda_n = \Lambda_n \setminus \Lambda_n^{\circ, sR}.$$

Every length- $m \leq s$ path generated by the range- R kernel and starting at $x \in \Lambda_n^{\circ, sR}$ stays in Λ_n . Therefore

$$\langle \delta_x, p(A_{R,n}) \delta_x \rangle = \langle \delta_0, p(A_R) \delta_0 \rangle, \quad x \in \Lambda_n^{\circ, sR}. \quad (5.8)$$

The radius here is sR , not R . Moreover,

$$\partial_{sR}\Lambda_n \subseteq \bigcup_{\|h\|_\infty \leq sR} (\Lambda_n \setminus (\Lambda_n - h)). \quad (5.9)$$

The union is finite and each summand has $o(|\Lambda_n|)$ elements by the Følner property. Hence $|\partial_{sR}\Lambda_n|/|\Lambda_n| \rightarrow 0$. With $K_p = \sup_{|t| \leq L+1} |p(t)|$, define

$$d_{n,R,p}(x) := |\langle \delta_x, p(A_{R,n})\delta_x \rangle - \langle \delta_0, p(A_R)\delta_0 \rangle|, \\ D_{n,R,p} := \frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} d_{n,R,p}(x).$$

Equations (5.8)–(5.9) give

$$D_{n,R,p} \leq 2K_p \frac{|\partial_{sR}\Lambda_n|}{|\Lambda_n|} \rightarrow 0. \quad (5.10)$$

Finally, the spectral theorem and Fourier transformation identify

$$\langle \delta_0, \phi(A)\delta_0 \rangle = \frac{1}{(2\pi)^d} \int_{[0,2\pi]^d} \phi(F(\theta)) d\theta = a_\phi.$$

Put

$$D_n^\phi := \frac{1}{|\Lambda_n|} \sum_{x \in \Lambda_n} |\langle \delta_x, \phi(A_n)\delta_x \rangle - a_\phi|.$$

For all sufficiently large n , functional calculus, (5.7), and (5.10) now give the single uniform error chain

$$\begin{aligned} D_n^\phi &\leq \|\phi(A_n) - p(A_n)\|_{\text{op}} \\ &\quad + \|p(A_n) - p(A_{R,n})\|_{\text{op}} \\ &\quad + D_{n,R,p} \\ &\quad + \|p(A_R) - p(A)\|_{\text{op}} \\ &\quad + \|p(A) - \phi(A)\|_{\text{op}} \\ &< 4\eta + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

This proves (5.5). $\square$

Lemma 5.2 (Gram upper bound for equiprobable pure states). *Let G be an $M \times M$ positive semidefinite Gram matrix with unit diagonal. Set*

$$d_j = (\sqrt{G})_{jj}, \quad q_j = \frac{d_j}{\text{tr } \sqrt{G}}, \quad u_j = \frac{1}{M}.$$

Then

$$\begin{aligned} \left(\frac{\text{tr } \sqrt{G}}{M} \right)^2 &\leq P_{\text{opt}}(G) \\ &\leq \left(\frac{\text{tr } \sqrt{G}}{M} \right)^2 + \sqrt{\lambda_{\max}(G)} \|q - u\|_1. \end{aligned} \quad (5.11)$$

Proof. The lower bound is (2.1). For $G \succ 0$, the upper bound is Theorem 1 of Ref. [2], rewritten in our normalization so that no prior-weight or hypothesis-count factor is implicit. That theorem assumes linearly independent states, equivalently a positive definite Gram matrix. For singular G , apply it to

$$G_\varepsilon = (1 - \varepsilon)G + \varepsilon I_M, \quad 0 < \varepsilon < 1.$$

Each G_ε is positive definite and has unit diagonal. As $\varepsilon \downarrow 0$, the positive square root, its trace and diagonal, $\lambda_{\max}$, and hence q all converge to their counterparts for G . Lemma 2.1 also gives $P_{\text{opt}}(G_\varepsilon) \rightarrow P_{\text{opt}}(G)$. Passing to the limit proves the singular case. $\square$

Theorem 5.3 (Exceptional-sector Gram transfer). *For each n , let G_n be an $M_n \times M_n$ positive semidefinite Gram matrix with unit diagonal. Let $\mathcal{B}_n$ be a set of m_n labels, let $G_{n,\mathcal{B}}$ be the associated principal submatrix, and let H_n be an $m_n \times m_n$ positive semidefinite matrix with unit diagonal. Suppose*

$$\frac{m_n}{M_n} \rightarrow 1, \quad (5.12)$$

$$\|G_{n,\mathcal{B}} - H_n\|_{\text{op}} \rightarrow 0, \quad (5.13)$$

$$\sup_n \|H_n\|_{\text{op}} < \infty, \quad (5.14)$$

and, for some $\alpha > 0$,

$$\frac{1}{m_n} \sum_{j=1}^{m_n} |(\sqrt{H_n})_{jj} - \alpha| \rightarrow 0. \quad (5.15)$$

Then the complete equiprobable ensemble satisfies

$$P_{\text{opt}}(G_n), P_{\text{SRM}}(G_n) \rightarrow \alpha^2. \quad (5.16)$$

Proof. Put $d_{n,j} = (\sqrt{H_n})_{jj}$ and $\bar{d}_n = m_n^{-1} \sum_j d_{n,j}$. Positivity and the unit diagonal give $0 \leq d_{n,j} \leq 1$, because

$$d_{n,j}^2 \leq \langle e_j, H_n e_j \rangle = 1.$$

Equation (5.15) implies $\bar{d}_n \rightarrow \alpha$ and

$$\frac{1}{m_n} \sum_j d_{n,j}^2 \rightarrow \alpha^2.$$

The latter is $P_{\text{SRM}}(H_n)$. To control the optimum, define

$$q_{n,j} = \frac{d_{n,j}}{m_n \bar{d}_n}, \quad u_{n,j} = \frac{1}{m_n}.$$

For all sufficiently large n , $\bar{d}_n > 0$, and

$$\begin{aligned} \|q_n - u_n\|_1 &= \frac{1}{\bar{d}_n} \frac{1}{m_n} \sum_j |d_{n,j} - \bar{d}_n| \\ &\leq \frac{1}{m_n \bar{d}_n} \sum_j |d_{n,j} - \alpha| + \frac{|\bar{d}_n - \alpha|}{\bar{d}_n} \rightarrow 0. \end{aligned}$$

Lemma 5.2, together with (5.14), now gives

$$P_{\text{opt}}(H_n) \longrightarrow \alpha^2.$$

By Lemma 2.1 and (5.13), the retained physical family has the same limits:

$$P_{\text{opt}}(G_{n,\mathcal{B}}), P_{\text{SRM}}(G_{n,\mathcal{B}}) \longrightarrow \alpha^2. \quad (5.17)$$

The same square-root perturbation bound also yields

$$\frac{\text{tr} \sqrt{G_{n,\mathcal{B}}}}{m_n} \longrightarrow \alpha. \quad (5.18)$$

Write $\beta_n = m_n/M_n$. Restricting a full-ensemble POVM to the retained labels and completing the resulting sub-POVM gives the upper bound below; extending an optimal retained-family POVM gives the lower bound:

$$\beta_n P_{\text{opt}}(G_{n,\mathcal{B}}) \leq P_{\text{opt}}(G_n) \leq \beta_n P_{\text{opt}}(G_{n,\mathcal{B}}) + (1 - \beta_n). \quad (5.19)$$

Equations (5.12), (5.17), and (5.19) prove $P_{\text{opt}}(G_n) \rightarrow \alpha^2$.

It remains to transfer the SRM. Put

$$t_n = \frac{\text{tr} \sqrt{G_n}}{M_n}.$$

Eigenvalue interlacing for the principal submatrix gives

$$t_n \geq \beta_n \frac{\text{tr} \sqrt{G_{n,\mathcal{B}}}}{m_n} \longrightarrow \alpha.$$

On the other hand, $t_n^2 \leq P_{\text{opt}}(G_n) \rightarrow \alpha^2$, so $t_n \rightarrow \alpha$. The universal squeeze

$$t_n^2 \leq P_{\text{SRM}}(G_n) \leq P_{\text{opt}}(G_n)$$

proves (5.16). $\square$

Corollary 5.4 (Følner comparison transfer). *Let $k \in \ell^1(\mathbb{Z}^d)$ be Hermitian with $k(0) = 1$, and suppose the symbol F of its convolution operator satisfies $0 < \mu \leq F \leq L < \infty$. Let Λ_n be a finite Følner sequence and H_n the compression of this convolution operator to $\ell^2(\Lambda_n)$. If G_n and retained label sets $\mathcal{B}_n$ satisfy*

$$\frac{|\mathcal{B}_n|}{M_n} \rightarrow 1, \quad \|G_n[\mathcal{B}_n] - H_n\|_{\text{op}} \rightarrow 0,$$

define

$$\alpha_F = \frac{1}{(2\pi)^d} \int_{[0,2\pi]^d} \sqrt{F(\theta)} d\theta. \quad (5.20)$$

Then

$$P_{\text{opt}}(G_n), P_{\text{SRM}}(G_n) \longrightarrow \alpha_F^2. \quad (5.21)$$

Proof. The matrices H_n have unit diagonal and $\|H_n\|_{\text{op}} \leq L$. Lemma 5.1, applied to $\phi(t) = \sqrt{t}$, gives (5.15) with

$$\alpha = \alpha_F > 0.$$

Theorem 5.3 applies. $\square$

5.2 The two-endpoint law

Theorem 5.5 (Unknown-length interval). *Under the uniform prior on $\mathcal{I}_n$,*

$$P_{\text{opt}}(G^{(n)}), P_{\text{SRM}}(G^{(n)}) \longrightarrow p_1(c)^2 \quad (5.22)$$

for every fixed $0 \leq c \leq 1$. The convergence is pointwise in fixed c . No finite-size rate or uniformity as $c \uparrow 1$ is claimed.

Proof. For $c = 0$, the hypotheses are orthogonal, whereas for $c = 1$ they are identical; both cases agree with the stated limit. Assume henceforth that $0 < c < 1$, and choose

$$L_n = \lceil \gamma \log n \rceil, \quad \gamma > |\log c|^{-1},$$

and retain only

$$\mathcal{B}_n = \{I \in \mathcal{I}_n : |I| \geq L_n\}.$$

The deleted prior mass is

$$\varepsilon_n = 1 - \frac{|\mathcal{B}_n|}{M_n} = O\left(\frac{\log n}{n}\right) \rightarrow 0. \quad (5.23)$$

On $\mathcal{B}_n$, define the comparison kernel

$$H_{I,J}^{(n)} = c^{|a-a'|+|b-b'|}. \quad (5.24)$$

It is the compression of the positive convolution kernel $c^{|u|+|v|}$. The two kernels agree for intersecting intervals. For disjoint intervals,

$$|G_{I,J} - H_{I,J}| \leq c^{|I|+|J|} \leq c^{2L_n}.$$

Consequently,

$$\|G_{\mathcal{B}_n} - H_{\mathcal{B}_n}\|_{\text{op}} \leq |\mathcal{B}_n| c^{2L_n} = o(1). \quad (5.25)$$

For completeness, the required Følner property follows directly. Under the endpoint identification $I = [a, b] \leftrightarrow (a, b)$,

$$\mathcal{B}_n = \{(a, b) \in \mathbb{Z}^2 : a \geq 1, b \leq n, b - a \geq L_n - 1\},$$

and

$$|\mathcal{B}_n| = \frac{(n - L_n + 1)(n - L_n + 2)}{2} \sim \frac{n^2}{2}. \quad (5.26)$$

Fix $h = (u, v) \in \mathbb{Z}^2$ and put $H = \|h\|_\infty$. If the indicators of $\mathcal{B}_n$ at $x = (a, b)$ and $x + h$ differ, at least one of the three affine inequalities defining $\mathcal{B}_n$ changes truth value. Such an x lies in a strip of respectively $|u|$, $|v|$, or $|v - u|$ lattice layers adjacent to $a = 1$, $b = n$, or $b - a = L_n - 1$. Because either x or $x + h$ belongs to $\mathcal{B}_n$, each layer contains at most $n + 2H + 1$ relevant points. Put $C_h = |u| + |v| + |v - u|$. Therefore

$$\frac{|(\mathcal{B}_n - h) \triangle \mathcal{B}_n|}{|\mathcal{B}_n|} \leq \frac{C_h(n + 2H + 1)}{|\mathcal{B}_n|} \rightarrow 0, \quad (5.27)$$

where (5.26) and $L_n = O(\log n) = o(n)$ were used. Thus $(\mathcal{B}_n)$ is a Følner sequence in $\mathbb{Z}^2$. The comparison convolution symbol is

$$F_c(\theta_1, \theta_2) = g_c(\theta_1)g_c(\theta_2),$$

with a uniform positive lower bound. The square-root spectral average in Corollary 5.4 factorizes as

$$\begin{aligned} \alpha_c &= \frac{1}{(2\pi)^2} \iint \sqrt{F_c} \\ &= \left[\frac{1}{2\pi} \int_0^{2\pi} \sqrt{g_c(\theta)} d\theta \right]^2 = p_1(c). \end{aligned} \quad (5.28)$$

The kernel $k(u, v) = c^{|u|+|v|}$ is Hermitian, belongs to $\ell^1(\mathbb{Z}^2)$, and has $k(0, 0) = 1$. Equations (5.23), (5.25), and (5.27) verify all remaining hypotheses of Corollary 5.4. That corollary transfers the comparison-kernel limit simultaneously to the optimum and SRM of the full, untruncated physical Gram family, giving $\alpha_c^2 = p_1(c)^2$. $\square$

6 Joint detection and exact localization

Add the no-anomaly state

$$|\Omega\rangle = |0\rangle^{\otimes n}$$

with fixed prior $\pi_0 \in (0, 1)$. No-change labels already appear in the permanent- and at-most-change formulations of Refs. [2, 10]; the result below concerns a returning interval and a no-change prior that remains nonzero as $n \rightarrow \infty$.

For known length, the complete joint prior is

$$\Pr(H_0) = \pi_0, \quad \Pr(a) = \frac{1 - \pi_0}{N}, \quad 1 \leq a \leq N. \quad (6.1)$$

For unknown length, it is

$$\Pr(H_0) = \pi_0, \quad \Pr(I) = \frac{1 - \pi_0}{M_n}, \quad I \in \mathcal{I}_n. \quad (6.2)$$

Thus the anomaly label is conditionally uniform given that an anomaly is present. The task is successful only if the measurement reports H_0 for $|\Omega\rangle$, or reports the exact anomalous interval otherwise. We write P_{joint}^* for the Bayes optimum over all POVMs. The results in this section concern this optimum; they do not assert asymptotic optimality of the weighted SRM for the augmented, nonuniform ensemble.

Theorem 6.1 (Fixed known length with no-anomaly prior). *Under the joint prior (6.1), for fixed i ,*

$$P_{\text{joint}}^*(N, i, r, \pi_0) \rightarrow \pi_0 + (1 - \pi_0)P_\infty(i, r). \quad (6.3)$$

If $i = i_n \rightarrow \infty$ and $N_n \rightarrow \infty$, then

$$P_{\text{joint}}^* \rightarrow \pi_0 + (1 - \pi_0)p_1(c^2). \quad (6.4)$$

Proof. If $r = 1$, every anomalous candidate equals $|\Omega\rangle$. The $N + 1$ labels are therefore represented by one physical state, with label priors π_0 and $(1 - \pi_0)/N$. Hence, exactly at finite N ,

$$P_{\text{joint}}^*(N, i, 1, \pi_0) = \max\left\{\pi_0, \frac{1 - \pi_0}{N}\right\} \rightarrow \pi_0. \quad (6.5)$$

This agrees with (6.3) because $P_\infty(i, 1) = 0$. Assume henceforth that $0 \leq r < 1$. Put $q = r^i$ and use

$$|\Phi_a^{(i)}\rangle = \sqrt{q}|\Omega\rangle + \sqrt{1 - q}|\xi_a^{(i)}\rangle.$$

If P_G^* and P_R^* are the equal-prior localization optima for the original and residual ensembles, respectively, then the explicit measurement that assigns $|\Omega\rangle$ to H_0 and uses an optimal residual measurement on $|\Omega\rangle^\perp$ gives

$$\pi_0 + (1 - \pi_0)(1 - q)P_R^* \leq P_{\text{joint}}^*.$$

Conversely, the no-anomaly contribution is at most π_0 , and completing the anomalous sub-POVM can only improve its conditional localization score, so

$$P_{\text{joint}}^* \leq \pi_0 + (1 - \pi_0)P_G^*.$$

For fixed i , Corollary 3.2 gives $(1 - r^i)P_R^* \rightarrow P_\infty(i, r)$, and the two bounds prove (6.3). In the growing-length regime, $|P_G^* - P_R^*| \leq r^{i/2}$ and $q \rightarrow 0$; Theorem 4.2 then proves (6.4). At $r = 0$, all $N + 1$ states are mutually orthogonal and $P_{\text{joint}}^* = 1$, also in agreement with both displayed limits. $\square$

Theorem 6.2 (Unknown length with no-anomaly prior). *Under the joint prior (6.2),*

$$P_{\text{joint}}^*(n, c, \pi_0) \rightarrow \pi_0 + (1 - \pi_0)p_1(c)^2. \quad (6.6)$$

Proof. Assume $0 < c < 1$ and use the long-interval family $\mathcal{B}_n$ from Theorem 5.5. For $I \in \mathcal{B}_n$, put

$$s_I = \langle \Omega | \Phi_I \rangle = c^{|I|} \leq c^{L_n} =: \eta_n \rightarrow 0$$

and

$$|\xi_I\rangle = \frac{|\Phi_I\rangle - s_I|\Omega\rangle}{\sqrt{1 - s_I^2}}.$$

To make the conditioning explicit, define the full-family anomalous localization optimum

$$P_{G,n}^{\text{all}} := \max_{\{M_I\}_{I \in \mathcal{I}_n}} \frac{1}{M_n} \sum_{I \in \mathcal{I}_n} \langle \Phi_I | M_I | \Phi_I \rangle = P_{\text{opt}}(G^{(n)}),$$

and define the two equal-prior optima on the retained family by

$$P_{G,n}^{\mathcal{B}} := \max_{\{M_I\}_{I \in \mathcal{B}_n}} \frac{1}{|\mathcal{B}_n|} \sum_{I \in \mathcal{B}_n} \langle \Phi_I | M_I | \Phi_I \rangle,$$

$$P_{R,n}^{\mathcal{B}} := \max_{\{M_I\}_{I \in \mathcal{B}_n}} \frac{1}{|\mathcal{B}_n|} \sum_{I \in \mathcal{B}_n} \langle \xi_I | M_I | \xi_I \rangle.$$

In each maximization the operators form a POVM on the support of the corresponding conditional ensemble, completed arbitrarily on its orthogonal complement. These are conditional localization probabilities: the factors $1 - \pi_0$ and $1 - \varepsilon_n$ are not included in their definitions. The pure-state trace distance gives the uniform comparison

$$|P_{R,n}^{\mathcal{B}} - P_{G,n}^{\mathcal{B}}| \leq \eta_n.$$

Use the optimal residual POVM for the long intervals in $|\Omega\rangle^\perp$, assign zero operators to deleted short intervals, and assign the remaining operator to H_0 . With $\varepsilon_n = 1 - |\mathcal{B}_n|/M_n$, this gives

$$P_{\text{joint}}^* \geq \pi_0 + (1 - \pi_0)(1 - \varepsilon_n)(1 - \eta_n^2)(P_{G,n}^{\mathcal{B}} - \eta_n).$$

The universal upper bound is

$$P_{\text{joint}}^* \leq \pi_0 + (1 - \pi_0)P_{G,n}^{\text{all}}.$$

Theorem 5.5 and $\varepsilon_n, \eta_n \rightarrow 0$ make the bounds coincide. If $c = 0$, all hypotheses are mutually orthogonal and the success probability is one. If $c = 1$, all hypotheses are identical and the optimum is $\max\{\pi_0, (1 - \pi_0)/M_n\} \rightarrow \pi_0$. $\square$

7 Numerical verification

The numerical calculations verify the finite-size Gram matrices and show how the asymptotic regimes are approached. None of the proofs relies on numerical evidence.

For small unknown-length instances, we solve the minimum-error problem in canonical Gram coordinates with both primal and dual semidefinite programs. The numerical grid covers $n = 3, \dots, 7$ and

$$c \in \{0.3, 0.6, 0.8, 0.9, 0.95, 0.99\}.$$

Raw solver objectives are retained as diagnostics but are not used directly as bounds. Let ρ_j denote the prior-weighted canonical states on their d_{supp} -dimensional support. For a numerical dual matrix Y , set

$$\delta_D = \max_j \max\{0, -\lambda_{\min}(Y - \rho_j)\}$$

and add the recorded scale-aware roundoff margin $\zeta_D > 0$. Then

$$Y_{\text{safe}} = Y + (\delta_D + \zeta_D)I, \quad U_{\text{safe}} = \text{tr } Y_{\text{safe}} \quad (7.1)$$

is rechecked against every dual PSD constraint. For exact matrices satisfying those constraints, it supplies a feasible upper bound.

For the primal matrices, project each raw M_j onto the PSD cone, write the result as $M_j^+ = B_j B_j^*$, and put $S = \sum_j M_j^+$. After verifying that $S \succ 0$ on the d_{supp} -dimensional canonical support, define

$$\widetilde{M}_j = S^{-1/2} M_j^+ S^{-1/2} = (S^{-1/2} B_j)(S^{-1/2} B_j)^*. \quad (7.2)$$

Then $\widetilde{M}_j \succeq 0$ and $\sum_j \widetilde{M}_j = I$ in exact arithmetic. In floating point, the implementation puts $T = \sum_j \widetilde{M}_j$, additionally contracts this sum by

$$\gamma = (\max\{1, \lambda_{\max}(T)\} + \zeta_P)^{-1}$$

and assigns the PSD remainder $I - \gamma T$ to one outcome. The resulting POVM $\{\widehat{M}_j\}$ is rechecked. In exact arithmetic, feasibility and weak duality give

$$L_{\text{safe}} = \sum_j \text{tr}(\rho_j \widehat{M}_j), \quad L_{\text{safe}} \leq P_{\text{opt}} \leq U_{\text{safe}}. \quad (7.3)$$

The implementation, however, uses IEEE double precision and conventional floating-point eigensolvers, without directed rounding, interval arithmetic, or exact-rational verification. We therefore report L_{safe} and U_{safe} as safeguarded floating-point bracket endpoints: the displayed inequality is rigorous conditional on exact feasibility of the postprocessed matrices, while the stored residuals quantify the numerical recheck at the working precision.

Across the 30 unknown-length cases, the largest raw absolute primal–dual gap is 6.782×10^{-9} , while the largest postprocessed floating-point bracket width $U_{\text{safe}} - L_{\text{safe}}$ is 8.854×10^{-9} . The largest rechecked completeness residual is 1.38×10^{-15} , and every rechecked primal and dual PSD violation is reported

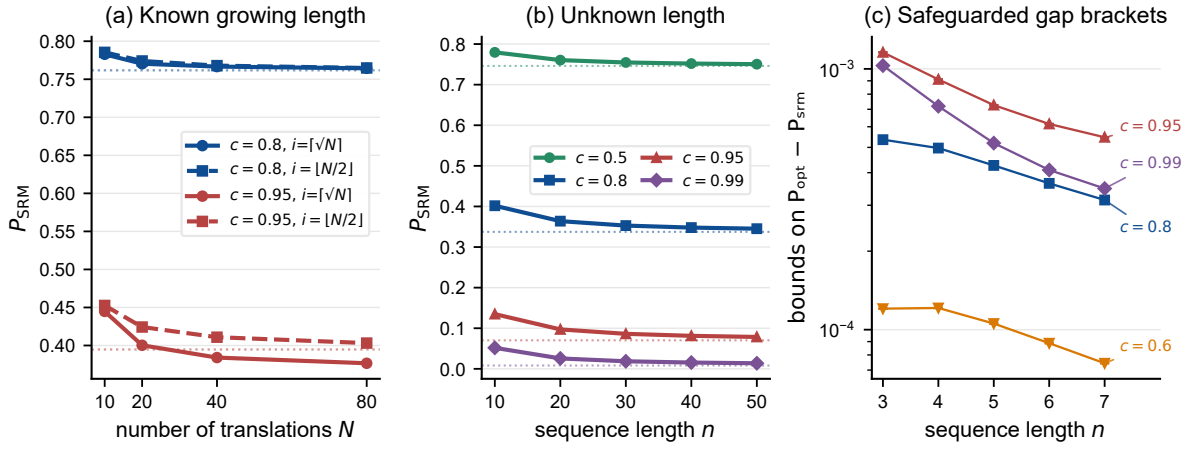


Figure 3: **Finite-size convergence and safeguarded floating-point SRM-gap brackets.** (a) SRM success for $i = \lceil \sqrt{N} \rceil$ and $i = \lfloor N/2 \rfloor$; dotted horizontal lines are $p_1(c^2)$. (b) Full dense physical-Gram SRM values through $n = 50$; dotted lines are $p_1(c^2)$, with high-overlap curves retained as stress tests. (c) Numerically rechecked endpoints $L_{\text{safe}} - P_{\text{SRM}}$ and $U_{\text{safe}} - P_{\text{SRM}}$ for $n \leq 7$. Conditional on exact feasibility, they bracket $P_{\text{opt}} - P_{\text{SRM}}$; here they are reported as safeguarded floating-point diagnostics. Plotting symbols mark only the midpoint of each displayed interval, not an estimate of $P_{\text{opt}} - P_{\text{SRM}}$; error bars are often smaller than the symbols.

as zero at the evaluator’s resolution. The smallest postprocessed primal eigenvalue and dual slack eigenvalue are respectively 3.4103×10^{-13} and 3.4102×10^{-13} . Full dense physical-Gram SRM computations for the single-interval model extend to $n = 50$, with candidate count $M_n = 1275$, over

$$c \in \{0.5, 0.8, 0.9, 0.95, 0.99\}.$$

For $n > 7$, only P_{SRM} is computed. The larger- n calculations therefore illustrate finite-size SRM convergence; they neither estimate $P_{\text{opt}} - P_{\text{SRM}}$ nor independently establish SRM optimality.

High-overlap cases converge slowly. The correlation length

$$\xi(c) = |\log c|^{-1}$$

is approximately 99.5 at $c = 0.99$, so even $n = 50$ has $n/\xi \approx 0.503$. The corresponding finite-size deviations should not be interpreted as asymptotic counterexamples.

The accompanying reproducibility package records the commands, package versions, file hashes, solver residuals, and data files. The environment is pinned to Python 3.12 and fixed versions of NumPy, SciPy, Matplotlib, CVXPY, Clarabel, and SCS; the reported clean-environment rerun used Python 3.12.10.

8 Discussion and limitations

Known and unknown lengths produce different analytic laws. A long known interval has one translation coordinate, and a unit shift changes two local factors; its elementary kernel parameter is therefore c^2 , giving $p_1(c^2)$. An unknown interval has two endpoint coordinates. In the long intersecting regime,

their symmetric-difference distance separates into the ℓ^1 distance of the endpoints, each contributing a kernel with parameter c ; the resulting law is $p_1(c)^2$. As labeled in Figure 2, these formulas correspond to distinct hypothesis sets and priors. Their juxtaposition expresses the one-coordinate/two-coordinate geometry rather than a same-prior value-of-information comparison.

For fixed known length, Theorem 3.1 proves the rigorous rate $P_{\text{opt}} - P_{\text{SRM}} = O_{i,r}(N^{-1/2})$. Exploratory finite-size calculations in some fixed-length parameter ranges are consistent with an N^{-1} decay, but the available computed sizes do not establish a uniform exponent. Determining whether the wrap-around boundary perturbation can be exploited to prove $O_{i,r}(N^{-1})$, or an asymptotic expansion with a nonzero N^{-1} coefficient, remains an open problem. By contrast, Theorems 4.2 and 5.5 are qualitative fixed-overlap limits: their proofs give neither a finite-size rate nor bounds uniform in a local regime $c = c_n \uparrow 1$.

The theorems provide collective-measurement benchmarks; an efficient physical implementation of the optimum remains open. Existing local protocols cover permanent changes and the $i = 1$ single-anomaly specialization [2, 12]. Matched local or sequential optima remain to be determined for fixed returning intervals with $i \geq 2$, growing known lengths, and the uniform-over-all-intervals ensemble. Further extensions include local asymptotic overlaps $c = c_n \rightarrow 1$, unknown or mixed anomalous states, nonuniform or minimax priors, restricted readout, and a growing number of disjoint intervals. Each changes the hypothesis geometry or admissible measurement class.

9 Conclusion

A returning quantum change retains enough translation structure for an exact asymptotic analysis. With fixed known length, the localization probability is a Toeplitz square-root integral; a growing known length gives the one-dimensional Toeplitz (single-change) functional evaluated at the effective compound overlap c^2 , namely $p_1(c^2)$, whereas an unknown length gives the two-endpoint law $p_1(c)^2$. The SRM reaches the global minimum-error limit in all three regimes. The exceptional-sector transfer theorem identifies the common mechanism carrying a Følner comparison limit to the full physical ensemble. Under the stated joint priors, a no-change hypothesis changes the optimal Bayes limit to $\pi_0 + (1 - \pi_0)L$; the augmented weighted SRM is not analyzed. Thus the contiguity constraint turns a general quantum multiple-hypothesis problem into tractable one- and two-dimensional spectral limits.

Data and code availability

This work is theoretical and generated no experimental data. All code and numerical data required to reproduce the results are available in the [public GitHub repository](#). The versioned release corresponding to this manuscript is [v1.1.0-paper1](#). An archival DOI will be added if that release is deposited in a permanent repository.

Author contributions

Xu Chen: Conceptualization; development of the physical model; formal analysis; mathematical derivation and proof of the theorems.

Xue Ma: Numerical computation; software implementation; generation of numerical data; data visualization and figure preparation.

Both authors contributed to the interpretation of the results, writing, and revision of the manuscript. The authors used OpenAI Codex to support literature and bibliographic search, structural revision, and language editing. It was also used for cross-checking software, numerical results, figures, and L^AT_EX. All authors reviewed the AI-assisted outputs and take full responsibility for the scientific content and the final manuscript.

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Competing interests

The authors declare no competing interests.

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